EXPERIMENT 1

Create and modify a Student database table using DDL commands. Demonstrate the use of CREATE, ALTER, TRUNCATE, and DROP commands with suitable examples.

Here are the Student table outputs for every step, in order, for your SQL experiment.

Step 1: CREATE TABLE

SQL:

```
CREATE TABLE Student (
    Student_ID INT PRIMARY KEY,
    Student_Name VARCHAR(50),
    Department VARCHAR(30),
    Age INT
);
```

Output – Empty Student table

Student_ID Student_Name Department Age

(empty)

Step 2: INSERT VALUES

SQL:

```
INSERT INTO Student VALUES
(101, 'Ravi', 'CSE', 20),
(102, 'Priya', 'ECE', 21),
(103, 'Arun', 'IT', 20);
```

Output – Student table

Student_ID	Student_Name	Department	Age
101	Ravi	CSE	20
102	Priya	ECE	21
103	Arun	IT	20

Step 3: ALTER TABLE

SQL:

ALTER TABLE Student

ADD Gender VARCHAR(50);

Output – After adding Gender

Student_ID	Student_Name	Department	Age	Gender
101	Ravi	CSE	20	NULL
102	Priya	ECE	21	NULL
103	Arun	IT	20	NULL

NULL means the Gender value has not been entered yet.

Step 4: UPDATE Gender

SQL:

UPDATE Student SET Gender = 'M'

WHERE Student_ID = 101;

UPDATE Student SET Gender = 'F'

WHERE Student_ID = 102;

UPDATE Student SET Gender = 'M'

WHERE Student_ID = 103;

Output – Updated Student table

Student_ID	Student_Name	Department	Age	Gender
101	Ravi	CSE	20	M
102	Priya	ECE	21	F
103	Arun	IT	20	M

Step 5: TRUNCATE TABLE

SQL:

TRUNCATE TABLE Student;

Output – Empty table

Student_ID Student_Name Department Age Gender

(no records)

The table structure remains, but all records are deleted.

Step 6: DROP TABLE

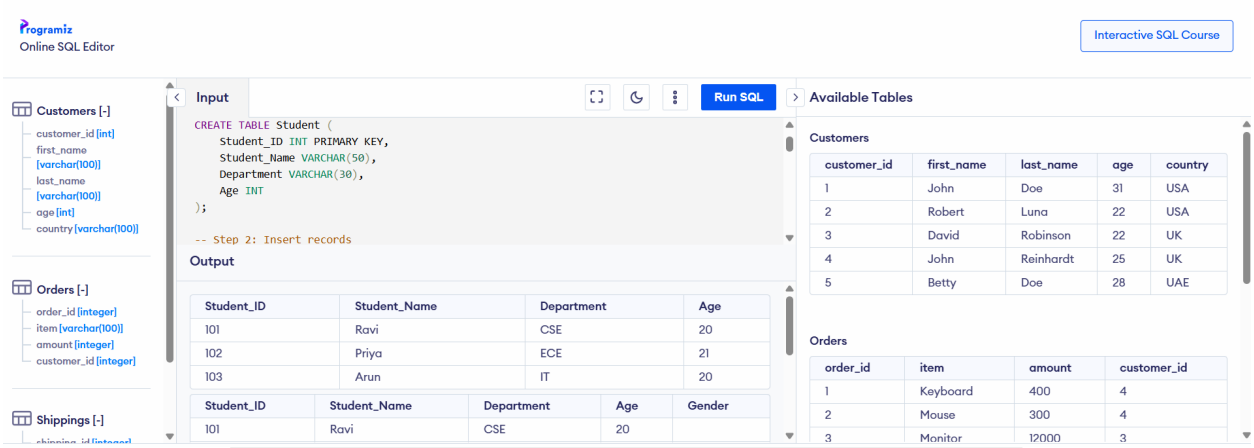
SQL:

DROP TABLE Student;

Output: The Student table is completely deleted. There is no table to display.

Final Result

The Student table was created, populated, modified using ALTER, cleared using TRUNCATE, and deleted using DROP. All four commands were demonstrated step by step.



Output

101	Ravi	CSE	20	
102	Priya	ECE	21	
103	Arun	IT	20	

Student_ID	Student_Name	Department	Age	Gender
101	Ravi	CSE	20	M
102	Priya	ECE	21	F
103	Arun	IT	20	M